

Three Attacks on Resolution Geometry

An Improved Stress-Test Battery, with Dimension Controls and One Concession

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Abstract

Three adversarial questions remain against the resolution-geometry (RG) results. (A) Is the lock an artifact of the observation protocol—same probes, same conclusions, nothing intrinsic? (B) Does the gluing homotopy *manufacture* the atlas, so that any blending would show the same holonomy? (C) Is the holonomy real but generic—would any smooth Grassmann bundle behave the same? The improved battery sharpens all three tests. Attack A fails: the lock is a dimension-free algebraic stratum, not a probe artifact. Across 800 random compatible inputs with varying dimensions and deliberate rank-deficient cases, the Schur, Fisher-pullback, and gauge-invariance identities hold to machine precision on the actual resolved-rank stratum. In the chaotic-vortex pipeline, five nondegenerate probe geometries recover the same structural stratum ($p = 6$, positive Schur floor, graded canonical spectrum, nonzero saturation gap), while a degenerate low-probe control correctly drops rank to $p = 4$. Attack B fails: holonomy is produced by curvature/noncommutativity, not by blending; a commuting family moves the subspace by 11.66 radians and yields holonomy 0.0003, whereas a non-commuting family moves it by 7.12 radians and yields holonomy 3.0500. Same-loop reparametrizations converge to the same value. Attack C is partly conceded: a generic smooth Grassmann bundle already has convergent nontrivial holonomy. What RG adds is not the existence of holonomy, but the admissibility and stratification underneath it: generic couplings rapidly cease to be Schur-admissible, whereas RG-built joint positive-definite geometries are admissible by construction, and the saturation counting law $\dim = p + q + pq$ holds across dimensions.

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1 The three attacks

Attack 1 (Protocol or dimension artifact). The lock and its conclusions are artifacts of the chosen observation protocol or dimension: *same probe geometry $\Rightarrow$ same conclusions*, with nothing intrinsic behind them.

Attack 2 (The homotopy manufactures the geometry). The gluing holonomy is created by the blending construction. A different embedding or a different blend would show the same atlas behavior, so the holonomy reports the homotopy, not any geometry.

Attack 3 (Not unique to RG). The holonomy is real but generic: any sufficiently smooth Grassmann bundle would show the same thing, so it is no evidence for RG specifically.

Each attack is answered by an improved battery (`ResolutionGeometryAttacks.battery_improved.py`, `numpy/scipy` only). We separate theorem-level algebraic identities from empirical structural persistence.

2 Attack A: is the lock a protocol or dimension artifact?

2.1 A1 — the lock is algebraic and dimension-free on the admissible stratum

The lock is not a numerical coincidence of one probe set; it is a chain of block-linear algebra identities. The correct theorem is not “all conclusions hold for every raw matrix”, but rather: for every compatible finite dimension, every positive-definite joint covariance, and the actual resolved rank stratum of the response, the lock identities hold. Rank-deficient protocols are not counterexamples; they are lower-stratum cases.

Theorem 2.1 (Dimension-free algebraic lock). *Let $O = R \oplus N$ be finite-dimensional and let*

$$\Sigma = \begin{pmatrix} \Sigma_R & C \\ C^\top & \Sigma_N \end{pmatrix}$$

be positive-definite with respect to this decomposition. Then the Schur complement

$$S_R = \Sigma_R - C \Sigma_N^{-1} C^\top$$

is positive-definite and

$$(R_b^\top \Sigma^{-1} R_b) = S_R^{-1}.$$

If $J = R_b A$ on the actual resolved-rank stratum, then

$$J^\top \Sigma^{-1} J = A^\top S_R^{-1} A.$$

Moreover, the canonical correlations of the pair (R, N) are invariant under independent orthogonal changes of resolved and null frames.

We tested the implementation on 800 random compatible inputs, with dimensions $m = 5, \dots, 24$, declared response dimensions $p = 1, \dots, 23$, and actual ranks $1, \dots, 23$, including deliberate rank-deficient cases.

lock identity	max error over 800 random inputs
block-inverse / Schur	5.29×10^{-14}
Fisher = pullback of inverse Schur	9.64×10^{-12}
gauge invariance of canonical correlations	6.46×10^{-15}
orthogonal gauge invariance of frame alignment	6.52×10^{-9}
rank-deficient cases detected correctly	0.00

Thus the algebraic lock is dimension-free under the stated positive-definiteness and rank assumptions. Dimension changes alter matrix sizes, spectrum lengths, and stratum counts; they do not break the identities.

2.2 A2 — the conclusions are structural; the values are protocol reports

We then ran the chaotic-vortex pipeline through deliberately dissimilar probe geometries, plus a degenerate low-probe control. The structural question is not whether the numbers are identical; they should not be. Different probes resolve different amounts. The question is whether the nondegenerate protocols identify the same stratum and whether insufficient protocols fail visibly rather than being hidden.

probe geometry	p	Schur _{min}	ρ_1	ρ_4	saturation gap
3 shells $r = 3, 4, 5$ (x6)	6	0.075	0.763	0.268	2.393
2 shells $r = 2.5, 6$ (x8)	6	0.076	0.850	0.395	6.201
random scatter (24)	6	0.074	0.774	0.239	1.802
one-sided arc (20)	6	0.074	0.486	0.008	0.035
far sparse $r = 8, 10$ (x5)	6	0.074	0.452	0.005	0.006
degenerate 2 probes (control)	4	—	—	—	—

Across every nondegenerate protocol, the recovered resolved rank is $p = 6$, the Schur floor is positive, the canonical spectrum is graded, and the saturation gap is nonzero. The far sparse and one-sided protocols report weak orientation content, as they should. The rank-degenerate control drops to $p = 4$ and therefore exposes insufficiency rather than manufacturing the conclusion.

Verdict 1. Attack A fails. The lock is algebraic and dimension-free on the admissible SPD/rank stratum. Probe geometry changes values, not the structural diagnosis; degenerate protocols correctly drop rank.

3 Attack B: does the homotopy manufacture the geometry?

3.1 B1 — holonomy is curvature/noncommutativity, not blending

If the homotopy manufactured the holonomy, then moving the resolved subspace “a lot” would always produce holonomy. It does not. We compare two two-parameter families of the resolved subspace built from skew generators Ω_1, Ω_2 : one with commuting controls ($\Omega_2 \parallel \Omega_1$), one with non-commuting controls.

controls	subspace travel around loop	holonomy
commuting	11.66 rad	0.0003
non-commuting	7.12 rad	3.0500

The commuting family drags the resolved subspace farther yet returns almost no holonomy; the non-commuting family moves it less and returns a large one. Holonomy measures curvature and noncommutativity of the loop, not the amount of blending.

3.2 B2 — holonomy is a loop invariant

Sampling the same loop uniformly versus clustered near corners gives the same limiting value:

samples N	uniform	clustered	diff
48	2.70867	2.70889	2.20×10^{-4}
96	2.70839	2.70845	5.50×10^{-5}
192	2.70832	2.70833	1.38×10^{-5}
384	2.70830	2.70831	3.44×10^{-6}

The value belongs to the loop, not to its parametrization. Different constructions that trace different loops may give different values, which is correct. The existence and geometric nature of holonomy is not manufactured by the sampling or blending.

Verdict 2. Attack B fails. Holonomy is produced by curvature/noncommutativity, not by blending, and it is stable under reparametrization of the same loop.

4 Attack C: is the phenomenon unique to RG?

This attack is partly conceded.

4.1 C1 — holonomy is generic (conceded)

A generic random smooth subspace bundle, represented as the image of $M_0 + sM_1 + tM_2 + stM_3$, already carries convergent, nontrivial holonomy:

samples N	overlap defect	holonomy
24	0.000	0.11924
48	0.000	0.12230
96	0.000	0.12413
192	0.000	0.12511

Thus RG did not invent holonomy. The holonomy itself is generic to smooth Grassmann bundles.

4.2 C2 — RG’s structure is not generic

What distinguishes an RG geometry from a generic smooth bundle is not the existence of holonomy, but the admissibility and stratification under the holonomy.

Schur admissibility. An RG geometry is a sub-block of a genuine positive-definite covariance, which forces the Schur floor and equivalently canonical correlations $\rho_i \leq 1$. A generic coupling need not satisfy this. Drawing random (Σ_R, C, Σ_N) with coupling scale α gives:

coupling scale α	0.3	0.7	1.2	2.0	RG-built
admissible (of 400)	252	44	2	0	400

Generic couplings rapidly leave the RG-admissible region; RG-built geometries from a joint positive-definite covariance are admissible by construction.

Saturation counting. The saturation counting law is dimension-dependent but exact on the tested top strata. The enlarged invariant set has Jacobian rank matching

$$\dim = p + q + pq.$$

(p, q)	invariant-rank	expected $p + q + pq$
(1, 1)	3	3
(2, 2)	8	8
(2, 3)	11	11
(3, 3)	15	15
(3, 4)	19	19

Verdict 3. Attack C is partly conceded. Holonomy is generic to smooth Grassmann bundles. RG’s specific content is the constrained admissible covariance geometry, the Schur/canonical-correlation strata, and the saturation counting law.

5 What survives

The attacks sharpen the claim. RG does not claim to have discovered holonomy, curvature, or parallel transport. It claims that observation covariance carries a specific resolved/null stratified geometry with algebraic admissibility constraints and completion invariants. The improved battery supports exactly that: the algebraic lock is dimension-free under SPD and rank assumptions; probe geometry changes values but not the nondegenerate stratum; holonomy is not a homotopy artifact; and RG-specific admissibility/stratification is not generic.

A Battery summary

All figures are from `Resolution_Geometry_Attacks_battery_improved.py`.

- A1: 800 random inputs, dimensions $m = 5..24$, declared $p = 1..23$, actual ranks 1..23; lock errors 5.29×10^{-14} , 9.64×10^{-12} , 6.46×10^{-15} , 6.52×10^{-9} .
- A2: five nondegenerate probe geometries recover $p = 6$, positive Schur floor, graded ρ , nonzero saturation gap; two-probe control drops to $p = 4$.
- B1: commuting controls travel 11.66 rad, holonomy 0.0003; non-commuting controls travel 7.12 rad, holonomy 3.0500.
- B2: same-loop uniform/clustered sampling difference decreases from 2.20×10^{-4} to 3.44×10^{-6} under refinement.
- C1: generic smooth Grassmann bundle has convergent nontrivial holonomy $0.11924 \rightarrow 0.12511$.
- C2: generic coupling admissibility 252/44/2/0 of 400 at $\alpha = 0.3/0.7/1.2/2.0$; RG-built 400/400 admissible. Saturation ranks match $p + q + pq$ for (1, 1), (2, 2), (2, 3), (3, 3), (3, 4).

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